BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, LAHORE

INTERMEDIATE (12TH CLASS) EXAMINATION, 2021

MATHEMATICS

Past Paper 2021

Retyped faithfully from the original board paper — same questions, no changes.

GROUP – I

Objective Type

Time Allowed: 30 Min. | **Maximum Marks:** 20

Note: Four possible answers A, B, C and D to each question are given. The choice which you think is correct, fill that circle in front of that question with Marker or Pen ink in the answer -book. Cutting or filling two or more circles will result in zero mark in that question.

Note: Objective Type

Note: (Group - I)

Q1. If f(x) = x/(x^2 - 4) then range of f(x) is:

(A) All real number
(B) Rational number
(C) All negative real number
(D) Integer

Q2. Lt (x->0) sin ax / sin bx =:

(A) b/a
(B) a
(C) a/b
(D) 1/b

Q3. If f(x) = 1/x^m then f'(x) = :

(A) -xm^-1
(B) -mx^-m-1
(C) -mx^-m+1
(D) -m^-1 x

Q4. The derivative of 1/(1+x) is:

(A) x
(B) 1 + x
(C) (1 + x)^-2
(D) -1(1+x)^-2

Q5. Integral sec x dx =:

(A) sec x + tan
(B) sec^2 x
(C) ln |sec x - tan x|
(D) ln |sec x + tan x| + c

Q6. Integral (2x+3)^1/2 dx =:

(A) (2x+3)^3/2 / 2 + c
(B) 1/3 (2x+3)^3/2 + c
(C) 1/2 (2x+3)^1/2 + c
(D) 1/3 (2x+3)^2 + c

Q7. Integral (a-2x)^3/2 dx =:

(A) 1/5 (a-2x)^3/2 + c
(B) 1/5 (a-2x)^5/2 + c
(C) -1/5 (a-2x)^5/2 + c
(D) -3/5 (a-2x)^5/2 + c

Q8. Midpoint of the line segment joining A (3, 1) and B (-2, -4) is:

(A) (1/2, 3/2)
(B) (5/2, 5/2)
(C) (1/2, 3/2)
(D) (1/2, 5/2)

Q9. Slope 'm' through A (x1, y1), B (x2, y2) is:

(A) (x2-x1)/(y2-y1)
(B) (x2+x1)/(y2-y1)
(C) (y2-y1)/(x2-x1)
(D) (y1-y2)/(x1+x2)

Q10. Distance between A (3, 1) and B (-2, -4) is:

(A) sqrt(17)
(B) 5 sqrt(2)
(C) sqrt(26)
(D) 2 sqrt(5)

Q11. x = 5 is the solution of inequality:

(A) 2x - 3 > 0
(B) 2x + 3 < 0
(C) x + 4 < 0
(D) x + 3 < 0

Q12. d/dx (cos^-1 x) =:

(A) 1/sqrt(1-x^2)
(B) -1/sqrt(1-x^2)
(C) 1/sqrt(1+x^2)
(D) 1/(1+x^2)

Q13. d/dx tan^-1 x =:

(A) 1/sqrt(1-x^2)
(B) 1/sqrt(1-x^2)
(C) 1/sqrt(1+x^2)
(D) 1/(1+x^2)

Q14. 1 + x + x^2/2! + x^3/3! + ---- is Maclaurin series for:

(A) e^x
(B) sqrt(1+x)
(C) cos x
(D) sin x

Q15. in equation of circle x^2 + y^2 = r^2 the centre of circle is:

(A) (x, y)
(B) (0, 0)
(C) (1, 0)
(D) (0, r)

Q16. In circle x^2 + y^2 + 2gx + 2fy + c = 0, the radius is:

(A) sqrt(g^2 + f^2 + c)
(B) g^2 + f^2 - c
(C) sqrt(g^2 + f^2 - c)
(D) g^2 + f^2 + c

Q17. The focus of parabola y^2 = 4ax is:

(A) (0, a)
(B) (-a, 0)
(C) (a, 0)
(D) (0, -a)

Q18. The vector PQ through P (0, 5) and Q (-1, -6) is:

(A) [-1, 11]
(B) [-1, -11]
(C) [0, 11]
(D) [1, 1]

Q19. Magnitude of vector u = 2i = 7i [sic, as printed] is:

(A) sqrt(53)
(B) sqrt(55)
(C) sqrt(48)
(D) sqrt(52)

Q20. In vectors a x b =:

(A) b x a
(B) -b x a
(C) -b
(D) -a x b

Subjective Type

Time Allowed: 2.30 Hrs. [sic, print 'Min.'] | **Maximum Marks:** 80

> Note: Subjective Type

> Note: (Group - I)

Q2. (Marks 16) Write short answer to any eight (8) question:

(i) Find the domain and range of the function g defined by : g (x) = sqrt(x^2 - 4)

(ii) The real valued functions f and g are given. Find fog (x), if f(x) = 3x^4 - 2x^2 and g (x) = 2/sqrt(x), x not= 0

(iii) Evaluate lim (theta->0) (1 - cos theta)/theta

(iv) Evaluate lim (x->1) (x^3 - 3x^2 + 2x - 1)/(x^3 - x)

(v) Find dy/dx if x^2 - 4xy - 5y = 0

(vi) Differentiate w.r.t. 'x' cot^-1 (x/a)

(vii) Find f'(x) if f(x) = sqrt(ln(e^2x + e^-3x))

(viii) Find y2 if x^3 + y^3 = a^3

(ix) Prove that d/dx(cosec^-1) = -1/(|x| sqrt(x^2 - 1))

(x) Differentiate (2x - 1)/sqrt(x^2 + 1)

(xi) Find the interval in which function is increasing or decreasing: f(x) = cos x, x in (-pi/2, pi/2)

(xii) Find y if y4 = sin 3x

Q3. (Marks 16) Write short answer to any eight (8) question:

(i) Use differentials to approximate the value of 4th-root(17)

(ii) Solve Integral dx/(sqrt(x+1) - sqrt(x))

(iii) Evaluate Integral cot(sqrt(x))/sqrt(x) dx

(iv) Solve Integral sec^2x / sqrt(tan x) dx

(v) Solve Integral e^x[1 - sin x + 2 cos x]dx

(vi) Evaluate Integral_0^4 sec x(sec x + tan x)dx

(vii) Solve the differential equation 1/x dy/dx = 1/2(1 + y^2)

(viii) Evaluate Integral x lnx dx

(ix) The points A (-5, -2), B(5, -4) are ends of a diameter of a circle. Find centre and radius of it.

(x) Transform the equation 5x - 12y + 39 = 0 into normal form.

(xi) Find k so that the lines joining A (7, 3), B(k, -6) and C(-4, 5), D(-6, 4) are parallel.

(xii) Find the lines represented by 2x^2 + 3xy - 5y^2 = 0

Q4. (Marks 18) Write short answer to any eight (8) question:

(i) Graph the inequality 5x - 4y <= 30

(ii) Find the equation of the circle with ends of diameter at (-3, 2) and (5, -6).

(iii) Find the centre of the circle 4x^2 + 4y^2 - 8x + 12y - 25 = 0

(iv) Find the length of the tangent from the point (-5, 10) to the circle 5x^2 + 5y^2 + 14x - 12y - 10 = 0

(v) Find the coordinates of the points of intersection of the line x+2y = 6 with the circle x^2 + y^2 - 2x - 2y - 39 = 0

(vi) Find the vertex of the parabola x^2 = 4 (y - 1)

(vii) Find the foci of the hyperbola x^2/16 - y^2/9 = 1

(viii) Find a unit vector in the direction of v = -sqrt(3)/2 i - 1/2 j

(ix) Find a vector whose magnitude is 4 and is parallel to 2i-3j + 6k

(x) If v is a vector for which v.i = 0, v.j = 0 and v.k = 0, find v

(xi) If a + b + c = 0, then prove that a x b = b x c = c x a

(xii) Find the volume of prallelepiped [sic, print for 'parallelepiped'] for which the vectors u = i - 4j-k, v = i - j - 2k and w = 2i - 3i + k [sic, as printed] are three edges.

(xiii) Give a force F = 2i + j - 3k acting at a point A (1, -2, 1). Find the moment of F about the point B(2, 0, -2)

GROUP – II

Objective Type

Time Allowed: 30 Min. | **Maximum Marks:** 20

Note: Four possible answers A, B, C and D to each question are given. The choice which you think is correct, fill that circle in front of that question with Marker or Pen ink in the answer -book. Cutting or filling two or more circles will result in zero mark in that question.

Note: Objective Type

Note: (Group-II)

Q1. The derivative of 1/(1+x) is:

(A) x
(B) 1 + x
(C) (1+x)^-2
(D) 1-(1+x)^-2

Q2. Integral cosx dx =:

(A) 1-sin^2x
(B) sqrt(1-sin^2x)
(C) sin x
(D) - sin x

Q3. Integral_1^2 (x^2+1)dx =:

(A) 10/3
(B) 3/10
(C) pi
(D) pi/2

Q4. If y = cot^-1x, then dy/dx =:

(A) 1/(1-x^2)
(B) -1/(1+x^2)
(C) 1
(D) 1

Q5. The derivative of ln (tanh x) is:

(A) 1/tanh x
(B) sech^2x/tanh x
(C) sech^2x
(D) sech x

Q6. x = at^2 and y = 2at are parametric equations of:

(A) Parabola
(B) Ellipse
(C) Circle
(D) Hyperbola

Q7. If y^2 + x^2 = a^2, then dy/dx =:

(A) x/y
(B) -y/x
(C) x/y
(D) y/x

Q8. The order of dy/dx = 4/3 x^3 + x - 3 is:

(A) 1
(B) 3/4
(C) 4/3
(D) -3

Q9. Integral_a^x 3x^2 dx =:

(A) x^3 + a^3
(B) x^3 - a^3
(C) 3x^3
(D) x^3

Q10. If theta is measured in radian then limit (theta->0) sin theta/theta =:

(A) 7
(B) 1/7
(C) 7pi/22
(D) 7pi/12

Q11. The measure of the angle between the lines ax^2 + 2hy + by^2 = 0 [sic, print for '2hxy'] is given by tan theta =:

(A) sqrt(h^2-ab)/(a-b)
(B) 2 sqrt(h^2-ab)/(a+b)
(C) (h^2-ab)/(a+b)
(D) infinity

Q12. If a = i - j and b = j + k then a.b =:

(A) 0
(B) 1
(C) -1
(D) 2

Q13. The feasible solution which maximize or minimize the objective function is called:

(A) Boundary
(B) Half plane
(C) Optimal solution
(D) Initial values

Q14. The value of c for [x^2/16 - y^2/49 = 1 is]:

(A) 16
(B) 49
(C) 65
(D) sqrt(65)

Q15. The equation of a straight line represented by x cos a + y sin a = P is called:

(A) Normal form
(B) Angular form
(C) Symmetric form
(D) P - form

Q16. The unit vector in the direction of v = [3, -4]:

(A) 5[3, -4]
(B) 1/5[3, -4]
(C) i
(D) j

Q17. The points A (-5, -2), B(5, -4) are ends point of a diameter of the circle. The centre will be:

(A) (0, 3)
(B) (0, -3)
(C) (5, 2)
(D) (-5, 4)

Q18. xy = 0 represents:

(A) A pair of lines
(B) Hyperbola
(C) Parabola
(D) Ellipse

Q19. The projections of v along u is:

(A) (u.v)/|u|
(B) (u.v)/|v|
(C) (u.v)/(|u||v|)
(D) (u.v)/(|u|+|v|)

Q20. An angle inscribed in a semi-circle is:

(A) 0
(B) pi/2
(C) pi
(D) 2pi

Subjective Type

Time Allowed: 2.30 Hrs. [sic, print 'Min.'] | **Maximum Marks:** 80

Note: Subjective Type

Note: (Group - II)

Q2. (Marks 16) Write short answer to any eight (8) question:

(i) Express the area A of a circle as a function of its circumference C.

(ii) For the real-valued function f(x) = (2x+1)/(2x-1), x > 1. Find f^-1(x)

(iii) Evaluate lim (x->3) (x-3)/(sqrt(x) - sqrt(3))

(iv) Find the domain and range of g(x) = |x -3|

(v) If y = (sqrt(x) - 1/sqrt(x))^2. find dy/dx

(vi) Find dy/dx if xy + y^2 = 2

(vii) Differentiate sin x w.r.t. cot x

(viii) Find dy/dx if y = x^2 ln(1/x)

(ix) Find y2 if y = x^2 . e^-x

(x) If y = ln (tanh x), find dy/dx

(xi) Find dy/dx if y = (x^2+5)(x^3+7)

(xii) Find f'(x) if f(x) = sqrt(ln(e^2x + e^-2x))

Q3. (Marks 16) Write short answer to any eight (8) question:

(i) Use differential to find dy/dx for xy + x = 4

(ii) Evaluate the integral Integral (3x+2)/sqrt(x) dx

(iii) Evaluate Integral (x+b)/(x^2+2bx+c)^1/2 dx

(iv) Evaluate Integral e^x (cos x + sin x) dx

(v) Evaluate Integral ((a-b)x)/((x-a)(x-b)) dx

(vi) Evaluate Integral (x^4 + 1)dx

(vii) Find the area above the x-axis and under the curve y = 5 - x^2 from x = -1 to x = 2.

(viii) Solve differential equation ydx + xdy = 0

(ix) Find mid-point of line segment joining A(-8, 3); B(2, -1)

(x) Two points 'P' and 'O'' given in xy-coordinate system. Find XY coordinates of 'P' referred to translated axis O'X and O'Y for P(-2, 6); O' (-3, 2).

(xi) Find equation of the line joining (-5, -3) and (9, -1)

(xii) Find equation of vertical line through (-5, 3).

Q4. (Marks 18) Write short answer to any eight (8) question:

(i) Graph the solution set of given linear inequality in xy-plane 2x + y <= 6.

(ii) Find the centre and radius of the circle with the given equation 5x^2 + 5y^2 + 14x + 12y - 10 = 0

(iii) Find the focus and vertex of the parabola x^2 = -16y

(iv) Write an equation of parabola with given elements: Focus (-3,1); directrix x - 2y - 3 = 0

(v) Find an equation of directrices of given hyperbola x^2/4 - y^2/9 = 1

(vi) Find the centre and eccentricity of given hyprbola [sic, print for 'hyperbola'] y^2/16 - x^2/9 = 1

(vii) Find the unit vector in the same direction as the vector v=[3,-4]

(viii) Find the constant a so that the vectors v = i - 3j + 4k and w = ai + 9j - 12k are parallel.

(ix) Find a vector of length 2 in the direction opposite that of v = -i + j + k

(x) Find the cosine of the angle theta between u and v = [2,-3,1] and u = [2, 4, 1]

(xi) Compute b x a. Check your answer by showing that b is perpendicular to b x a : a = 2i + j - k; b = i - j + k.

(xii) If a + b + c = 0, then prove that a x b = b x c = c x a

(xiii) Give a force F = 2i + j - 3k acting at a point A(1, -2, 1). Find the moment of F about the point B(2,0,-2).

> Note: PART - II

> Note: Note: Attempt any THREE questions.

Q5. (a) Find value of k, if the function f(x) = {(sqrt(2x+5) - sqrt(x+7))/(x-2), x not=2; k, x=2} is continuous at x = 2 (5)

(a) Find value of k, if the function f(x) = {(sqrt(2x+5) - sqrt(x+7))/(x-2), x not=2; k, x=2} is continuous at x = 2 (5)

(b) If y = tan(p tan^-1 x) then show that (1+x^2)y1 - p(1+y^2) = 0 (5)

Q6. (a) Evaluate Integral sqrt(2)/(sin x + cos x) dx (5)

(a) Evaluate Integral sqrt(2)/(sin x + cos x) dx (5)

(b) Find the equations of the through the intersection of the lines x - y - 4 = 0 and 7x + y + 20 = 0 and parallel to the line 6x + y - 14 = 0 (5)

Q7. (a) Find the area bounded by the curve y = x^3 - 4x and the x - axis. (5)

(a) Find the area bounded by the curve y = x^3 - 4x and the x - axis. (5)

(b) Maximize f(x,y) = 2x + 5y subject to the constraints 2y - x <= 8, x - y <= 4, x >= 0, y >= 0 (5)

Q8. (a) Write equation of the circle passing through the points A(-7, 7), B(5, -1) and C (10,0) (5)

(a) Write equation of the circle passing through the points A(-7, 7), B(5, -1) and C (10,0) (5)

(b) Find a vector of length 5 in the direction opposite that of v = i-2j+3k (5)

Q9. (a) Show that y = (ln x)/x has maximum value at x = e (5)

(a) Show that y = (ln x)/x has maximum value at x = e (5)

(b) Find focus, vertex and directrix of parabola x^2 - 4x - 8y + 4 = 0 (5)